Day 02 – Two-LAN Network Configuration and Inter-Network Routing

Project Category: Network Engineering
Tool: Cisco Packet Tracer
Difficulty: Beginner
Status: Completed
Focus Areas: LAN Configuration, IPv4 Addressing, Default Gateways, Routing and Network Troubleshooting

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1. Project Overview

This project involved designing and configuring a small network environment consisting of two separate Local Area Networks (LANs): Administration and Technical Support.

Each department was connected to its own access switch, with both switches connected to a central Cisco router.

The router provided communication between the two networks through its separate Gigabit Ethernet interfaces.

The project focused on understanding basic network segmentation, IPv4 addressing, default gateways, directly connected routes and inter-network communication.

2. Project Objectives

The main objectives were to:

- Design and implement two separate LANs.
- Connect two switches to a central router.
- Configure router interfaces with static IPv4 addresses.
- Assign static IP addresses to end devices.
- Configure appropriate subnet masks and default gateways.
- Establish communication within and between the LANs.
- Verify connectivity using ICMP ping tests.
- Inspect the router's routing table.
- Identify and troubleshoot an incorrect default gateway configuration.

3. Network Topology

The network consisted of:

Device	Quantity
Cisco 2911 Router	1
Cisco 2960 Switches	2
End-user PCs	4
Copper Straight-Through Cables	6

Physical connections

- R1 GigabitEthernet0/0 → SW1 GigabitEthernet0/1
- R1 GigabitEthernet0/1 → SW2 GigabitEthernet0/1
- SW1 FastEthernet0/1 → PC1
- SW1 FastEthernet0/2 → PC2
- SW2 FastEthernet0/1 → PC3
- SW2 FastEthernet0/2 → PC4

4. IP Addressing Scheme

Two separate /24 networks were implemented.

LAN 1 – Administration

Network address: `192.168.10.0/24`

Device	IP Address	Subnet Mask	Default Gateway
R1 G0/0	192.168.10.1	255.255.255.0	N/A
PC1	192.168.10.10	255.255.255.0	192.168.10.1
PC2	192.168.10.11	255.255.255.0	192.168.10.1

LAN 2 – Technical Support

Network address: `192.168.20.0/24`

Device	IP Address	Subnet Mask	Default Gateway
R1 G0/1	192.168.20.1	255.255.255.0	N/A
PC3	192.168.20.10	255.255.255.0	192.168.20.1
PC4	192.168.20.11	255.255.255.0	192.168.20.1

5. Router Configuration

The Cisco 2911 router was configured with a hostname and two active Gigabit Ethernet interfaces.

Initial configuration

```
None
enable
configure terminal
hostname R1
```

Administration interface

```
None
interface gigabitEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

Technical Support interface

```
None
interface gigabitEthernet 0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

Saving the configuration

```
None
end
write memory
```

The `no shutdown` command enabled both router interfaces.

6. End-Device Configuration

All four PCs were assigned static IPv4 configurations through:

PC → Desktop → IP Configuration → Static

Each PC received:

- A unique IPv4 address.
- A subnet mask of `255.255.255.0`.
- The IP address of its connected router interface as its default gateway.

No DNS configuration was required because the project focused on IP-based connectivity rather than domain-name resolution.

7. Router Interface Verification

The following command was executed:

```
None
show ip interface brief
```

The output confirmed:

Interface	IP Address	Status	Protocol
GigabitEthernet0/0	192.168.10.1	Up	Up
GigabitEthernet0/1	192.168.20.1	Up	Up

Both interfaces were operational.

8. Routing Table Verification

The router's routing table was inspected using:

```
None
show ip route
```

The following directly connected routes were confirmed:

```
None
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0
C 192.168.20.0/24 is directly connected, GigabitEthernet0/1
```

The router also displayed local routes for its own interface addresses.

This confirmed that R1 had learned both networks as directly connected routes and could route traffic between them.

9. Connectivity Testing

Connectivity was tested from PC1 using the Command Prompt.

Test 1 – Same-LAN communication

```
None
ping 192.168.10.11
```

Destination: PC2

Result: 4 packets received, 0% packet loss.

Test 2 – Communication between LANs

```
None
ping 192.168.20.10
```

Destination: PC3

The initial attempt experienced one timeout, but the subsequent test returned four successful replies with 0% packet loss.

Test 3 – Communication with another device in LAN 2

```
None
ping 192.168.20.11
```

Destination: PC4

The initial attempt experienced one timeout, but the subsequent test returned four successful replies with 0% packet loss.

These tests demonstrated successful communication within the local network and across the two separate networks.

10. Troubleshooting Exercise

A deliberate configuration error was introduced on PC4 to simulate a common network connectivity problem.

Identified fault

PC4's default gateway was changed from:

```
None
192.168.20.1
```

to:

```
None
192.168.20.254
```

Observed behaviour

A connectivity test from PC1 to PC4 was performed:

```
None
ping 192.168.20.11
```

The test returned repeated request timeouts.

Root cause

PC4 had an incorrect default gateway.

Although PC4 could receive traffic from another network, it could not correctly forward replies to destinations outside its local subnet because the configured gateway address was not assigned to the router.

Corrective action

The default gateway on PC4 was restored to:

```
None
192.168.20.1
```

The connectivity test was repeated.

Final result

Four packets were sent and four packets were received, resulting in 0% packet loss.

The fault was successfully identified and resolved.

11. Skills and Concepts Demonstrated

This practical exercise strengthened my understanding of:

- Local Area Network design.
- IPv4 addressing and subnet masks.
- Default gateway configuration.
- Router interface configuration.
- Directly connected routes.
- Basic inter-network routing.
- ICMP connectivity testing.
- ARP and initial ping behaviour.
- Network troubleshooting and fault isolation.
- Cisco IOS command-line operations.

12. Conclusion

This project successfully demonstrated how two separate departmental LANs can communicate through a central router.

By configuring router interfaces, assigning static IP addresses, setting default gateways and verifying connectivity, I gained practical experience with fundamental networking concepts.

The troubleshooting exercise also demonstrated the importance of correct default gateway configuration and how a seemingly small addressing error can interrupt communication between networks.

This project forms part of my ongoing practical Network Engineering and CCNA learning journey.

13. Project Deliverables

The completed project includes:

- Cisco Packet Tracer topology file.
- Network topology screenshots.
- Router configuration and verification evidence.
- End-device IP addressing records.
- Connectivity testing results.
- Troubleshooting evidence.
- Technical project documentation.

Project Status: Completed.